SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - Industry based Problem-Solving Question

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4**	**Assessment:**	**Assessment Tool 2**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	To apply Statistical inferences such as testing of Hypothesis and Point estimates		
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Section / Batch:	2024	**Date:**	19-08-26

Code of Conduct

I __**M Poojitha Reddy - 192472352**_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __**M Poojitha Reddy**________________________

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

1. CYBERSECURITY — Hypothesis Testing Using p-value

Problem

A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate using a p-value.

1. Population and Sample

Item	Description
Population	All malicious attacks that the cybersecurity system may encounter
Sample	200 randomly selected attacks
Successful detections	194
Failed detections	6

2. Parameter of Interest

Let **p** = true detection rate of the new cybersecurity algorithm.

Historical detection rate:

$\mathbf{p_0 = 0.95}$

3. Point Estimate

The sample detection rate is:

$\mathbf{\hat{p} = x/n}$

$\mathbf{\hat{p} = 194/200 = 0.97}$

Therefore, the new algorithm has a sample detection rate of:

$\hat{p} = 97\%$

The observed improvement is:

$$97\% - 95\% = 2 \text{ percentage points}$$

4. Hypotheses

The question asks whether the algorithm has **improved** the detection rate.

$H_0: p = 0.95$

$H_1: p > 0.95$

This is a **right-tailed test**.

5. Significance Level

Unless otherwise specified, take:

$\alpha = 0.05$

6. Test Statistic

For a one-sample proportion test:

$z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0)/n]}$

Substituting:

$z = (0.97 - 0.95) / \sqrt{[(0.95)(0.05)/200]}$

$z = 0.02 / 0.01541$

$z \approx 1.30$

7. p-value

For a right-tailed test:

p-value = P(Z > 1.30)

From the standard normal distribution:

p-value ≈ 0.097

8. Decision

Comparison	Result
p-value	0.097
α	0.05
Comparison	0.097 > 0.05
Decision	**Fail to reject H_0**

Decision Map

p-value = 0.097

↓

0.097 > 0.05

↓

Fail to reject H_0

↓

Improvement is NOT statistically significant

9. Interpretation

Although the sample detection rate increased from **95% to 97%**, the p-value is greater than the 5% significance level. Therefore, the observed 2 percentage-point improvement could reasonably be due to random sampling variation.

10. Industry-Oriented Conclusion

There is insufficient statistical evidence at the 5% significance level to conclude that the new cybersecurity algorithm has significantly improved the malicious-activity detection rate. Although the new algorithm achieved a higher sample detection rate of 97%, the improvement from 95% is not statistically significant based on this sample.

Final Answer: The new algorithm cannot yet be claimed to have significantly improved detection performance.

2. HEALTHCARE — Hypothesis Testing Using p-value

Problem

A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with standard deviation 12 minutes. Test whether the new system significantly reduces waiting time using the p-value approach.

1. Population and Sample

Item	Description
Population	All patients using the hospital appointment system
Sample	64 patients after implementation
Sample mean	38 minutes
Sample standard deviation	12 minutes
Historical mean	42 minutes

2. Parameter of Interest

Let μ = true average patient waiting time after implementation.

Historical value:

μ_0 = 42 minutes

3. Point Estimate

The point estimate of the population mean is the sample mean:

$\bar{x}$ = 38 minutes

Therefore:

Estimated average waiting time = 38 minutes

Observed reduction:

42 − 38 = 4 minutes

4. Hypotheses

The hospital wants to determine whether waiting time has **reduced**.

H_0: $\mu = 42$ minutes

H_1: $\mu < 42$ minutes

This is a **left-tailed test**.

5. Significance Level

$\alpha = 0.05$

6. Test Statistic

Since the population standard deviation is unknown, use a **one-sample t-test**.

Formula:

$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$

Given:

$\bar{x} = 38$
$\mu_0 = 42$
$s = 12$
$n = 64$

Therefore:

$t = (38 - 42)/(12/\sqrt{64})$

$$\mathbf{t = -4/(12/8)}$$

$$\mathbf{t = -4/1.5}$$

$$\mathbf{t \approx -2.67}$$

Degrees of freedom:

$$\mathbf{df = n - 1 = 63}$$

7. p-value

For:

$$\mathbf{t = -2.67,\ df = 63}$$

The left-tailed p-value is approximately:

$$\mathbf{p\text{-value} \approx 0.0049}$$

8. Decision

Item	Value
p-value	0.0049
α	0.05
Comparison	$0.0049 < 0.05$
Decision	**Reject H_0**

Decision Map

p-value = 0.0049

↓

0.0049 < 0.05

↓

Reject H_0

↓

Waiting-time reduction is statistically significant

9. Interpretation

The p-value is approximately **0.0049**, which is less than 0.05. Therefore, there is strong statistical evidence that the average patient waiting time after introducing the new system is lower than 42 minutes. The sample shows a reduction of:

$$42 - 38 = 4 \text{ minutes}$$

10. Industry-Oriented Conclusion

The hospital has sufficient statistical evidence at the 5% significance level to conclude that the new appointment scheduling system significantly reduces average patient waiting time. The observed average waiting time decreased from 42 minutes to 38 minutes, representing an average reduction of 4 minutes.

Final Answer: The new scheduling system significantly improves patient waiting time.

Optional 95% Confidence Interval

Approximate 95% CI:

$$\bar{x} \pm t_{0.975,63}(s/\sqrt{n})$$

Using t ≈ 2.00:

38 ± 2(1.5)

= 38 ± 3

Therefore:

$$\mathbf{95\%\ CI \approx (35, 41)\ minutes}$$

Since 42 minutes is outside this interval, the confidence interval also supports a significant reduction.

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3. SOCIAL MEDIA ANALYTICS — Hypothesis Testing Using p-value

Problem

A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with standard deviation 0.8 hours. Determine whether the data provide sufficient evidence to reject the company's claim at $\alpha = 0.05$.

1. Population and Sample

Item	Description
Population	All users of the social media platform
Sample	100 randomly selected users
Sample mean	3.2 hours
Sample standard deviation	0.8 hours
Claimed mean	3 hours

2. Parameter of Interest

Let **μ** = true average daily usage time of all users.

Claimed value:

μ_0 = 3 hours

3. Point Estimate

The sample mean is:

$\bar{x}$ = 3.2 hours

Therefore, the estimated average daily usage is **3.2 hours**.

Difference from claimed value:

$3.2 - 3.0 = 0.2$ hours

4. Hypotheses

Since the question asks whether the data provide evidence to **reject the claim**, use a two-tailed test:

H_0: $\mu = 3$ hours

H_1: $\mu \neq 3$ hours

5. Significance Level

Given:

$\alpha = 0.05$

6. Test Statistic

Using the one-sample test statistic:

$$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$$

Substitute:

$$t = (3.2 - 3)/(0.8/\sqrt{100})$$

$$t = 0.2/0.08$$

$$t = 2.50$$

With:

$$df = 100 - 1 = 99$$

For a two-tailed test, the p-value is approximately:

$$\text{p-value} \approx 0.014$$

7. p-value

p-value ≈ 0.014

This is the probability of observing a difference at least as extreme as the sample result if the true average usage time were actually 3 hours.

8. Decision

Item	Value
p-value	≈ 0.014
α	0.05
Comparison	$0.014 < 0.05$
Decision	**Reject H_0**

Decision Map

Sample mean = 3.2 hours

↓

Test statistic = 2.50

↓

p-value ≈ 0.014

↓

$0.014 < 0.05$

↓

Reject H_0

9. Interpretation

Since the p-value is less than 0.05, there is sufficient statistical evidence to conclude that the true average daily usage time is **different from 3 hours**.

The sample mean is 3.2 hours, which is higher than the claimed value.

10. Industry-Oriented Conclusion

At the 5% significance level, there is sufficient evidence to reject the company's claim that the average daily usage time is 3 hours. The data suggest that the actual average usage time is different from 3 hours, with the sample indicating a higher average of 3.2 hours.

Final Answer: The company's 3-hour claim should be rejected at the 5% significance level.

95% Confidence Interval

Approximate 95% CI:

$3.2 \pm 1.96(0.8/\sqrt{100})$

$= 3.2 \pm 1.96(0.08)$

$= 3.2 \pm 0.157$

Therefore:

$$\textbf{95\% CI} \approx (3.043, 3.357) \textbf{ hours}$$

Since 3 hours is not inside the confidence interval, the confidence interval also supports rejecting the company's claim.

4. MACHINE LEARNING — Hypothesis Testing Using p-value

Problem

A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations. Formulate suitable null and alternative hypotheses and determine whether the improvement is statistically significant using a p-value.

1. Population and Sample

Item	Existing Model	New Model
Population	Classification tasks	Classification tasks
Sample size	—	500
Accuracy	82%	86%
Correct predictions	—	430

For the new model:

$\mathbf{x = 0.86 \times 500 = 430}$

2. Parameter of Interest

Let **p** = true classification accuracy of the new model.

Existing model accuracy:

$\mathbf{p_0 = 0.82}$

3. Point Estimate

The sample accuracy is:

$\mathbf{\hat{p} = 430/500 = 0.86}$

Therefore:

$\mathbf{\hat{p} = 86\%}$

Observed improvement:

86% − 82% = 4 percentage points

4. Hypotheses

The question asks whether the new model is **better**.

Therefore:

$$H_0: p = 0.82$$

$$H_1: p > 0.82$$

This is a **right-tailed test**.

5. Significance Level

Use:

$\alpha = 0.05$

6. Test Statistic

For a one-sample proportion test:

$z = (\hat{p} - p_0)/\sqrt{[p_0(1 - p_0)/n]}$

Substitute:

$z = (0.86 - 0.82)/\sqrt{[(0.82)(0.18)/500]}$

$z = 0.04/0.01718$

$z \approx 2.33$

7. p-value

For a right-tailed test:

p-value = $P(Z > 2.33)$

Approximately:

p-value ≈ 0.010

8. Decision

Item	Value
Existing accuracy	82%
New accuracy	86%
Improvement	4 percentage points

Item	Value
p-value	≈ 0.010
α	0.05
Decision	**Reject H_0**

Since:

$$\mathbf{0.010 < 0.05}$$

we reject the null hypothesis.

9. Interpretation

The p-value of approximately 0.010 indicates that, if the true accuracy were only 82%, there would be about a **1% probability** of observing an accuracy of 86% or higher due to random sampling variation.

Since this probability is small, the observed improvement provides statistically significant evidence against the null hypothesis.

10. Industry-Oriented Conclusion

At the 5% significance level, there is sufficient statistical evidence to conclude that the new classification model performs significantly better than the existing model. The new model achieved 86% accuracy compared with 82%, an improvement of 4 percentage points, and this improvement is statistically significant.

Final Answer: The new model's improvement is statistically significant at the 5% level.

ML Decision Map

Existing accuracy = 82%

↓

New accuracy = 86%

↓

Improvement = 4%

↓

p-value ≈ 0.010

↓

0.010 < 0.05

↓

Reject H_0

↓

New model performs significantly better ✅

5.MANUFACTURING / QUALITY CONTROL — CI + p-value

Problem

A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective. Perform a hypothesis test at the 5% significance level using both:

(a) Confidence interval approach

(b) p-value approach

Compare the conclusions.

1. Population and Sample

Item	Value
Population	All products manufactured
Sample size	2,000
Defective products	55
Non-defective products	1,945
Claimed defect rate	2%
Significance level	5%

2. Parameter of Interest

Let:

p = true defect rate of the manufacturing process

The manufacturer's claim is:

$p_0 = 0.02$

3. Point Estimate

The sample defect rate is:

$\hat{p} = x/n$

$\hat{p} = 55/2000$

$\hat{p} = 0.0275$

Therefore:

$\hat{p} = 2.75\%$

Observed defect rate is therefore **2.75%**, which is higher than the claimed 2%.

Difference:

2.75% − 2% = 0.75 percentage points

(a) CONFIDENCE INTERVAL APPROACH

4. Hypotheses

Because the manufacturer claims the defect rate is **only 2%**, we are interested in whether the actual defect rate is higher:

$H_0: p = 0.02$

$H_1: p > 0.02$

This is a **right-tailed test**.

For the confidence-interval method, we use the standard **95% two-sided confidence interval**.

5. 95% Confidence Interval

Formula:

$$\mathbf{CI = \hat{p} \pm z_{0.975} \sqrt{[\hat{p}(1 - \hat{p})/n]}}$$

For 95% confidence:

$$\mathbf{z_{0.975} = 1.96}$$

Substitute:

$$\mathbf{CI = 0.0275 \pm 1.96\sqrt{[(0.0275)(0.9725)/2000]}}$$

Standard error:

$$\mathbf{SE \approx 0.00366}$$

Margin of error:

$$\mathbf{ME = 1.96 \times 0.00366}$$

$$\mathbf{ME \approx 0.00717}$$

Therefore:

$$\mathbf{CI = 0.0275 \pm 0.00717}$$

95% CI ≈ (0.0203, 0.0347)

In percentage form:

95% CI ≈ (2.03%, 3.47%)

6. Confidence Interval Interpretation

The 95% confidence interval for the true defect rate is approximately:

(2.03%, 3.47%)

The claimed defect rate is:

2%

Notice that **2% is below the lower confidence limit of approximately 2.03%**.

Therefore, the confidence interval provides evidence that the actual defect rate is higher than the claimed 2%.

CI Decision Map

95% CI = (2.03%, 3.47%)

↓

Claimed rate = 2%

↓

2% is outside the interval

↓

Evidence against 2% claim

↓

Reject H_0

(b) p-VALUE APPROACH

7. Test Statistic

For a one-sample proportion test:

$$z = (\hat{p} - p_0)/\sqrt{[p_0(1 - p_0)/n]}$$

Substitute:

$$z = (0.0275 - 0.02)/\sqrt{[(0.02)(0.98)/2000]}$$

$$z = 0.0075/0.00313$$

$$z \approx 2.40$$

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8. p-value

Since this is a right-tailed test:

p-value = $P(Z > 2.40)$

Approximately:

p-value ≈ 0.008

9. Compare p-value with α

Item	Value
p-value	≈ 0.008
α	0.05
Comparison	$0.008 < 0.05$
Decision	**Reject H_0**

Since:

p-value < α

we reject the null hypothesis.

10. Interpretation

The p-value of approximately **0.008** indicates that if the true defect rate were actually 2%, the probability of obtaining a sample defect rate as high as 2.75% or higher would be less than 1%.

This provides strong evidence against the manufacturer's claim.

COMPARISON OF BOTH APPROACHES

Method	Result	Decision	Conclusion
95% Confidence Interval	(2.03%, 3.47%)	Reject H_0	2% claim is not supported
p-value	≈ 0.008	Reject H_0	Significant evidence of higher defect rate
Overall	Both agree	**Reject H_0**	Actual defect rate is likely > 2%

Industry-Oriented Conclusion

The sample defect rate is **2.75%**, which is higher than the manufacturer's claimed rate of 2%.

Both statistical approaches lead to the same conclusion:

- The **95% confidence interval** is approximately **2.03% to 3.47%**, which places the claimed 2% rate below the interval.
- The **p-value is approximately 0.008**, which is less than the 5% significance level.

Therefore, there is **sufficient statistical evidence to conclude that the true defect rate is greater than 2%**.

The manufacturer should **reject the 2% defect-rate claim and investigate the production process** for possible quality-control issues.

Final Answer: The production process has a statistically significant defect rate above 2% at the 5% significance level.